Do You Know?

- ❖ The north pole of a magnet points roughly toward Earth's north pole and vice-versa. That's because Earth itself contains magnetic materials and behaves like a gigantic magnet.
- ❖ If you cut a bar magnet in half, it's a bit like cutting an earthworm in half! You get two brand new, smaller magnets, each with its own north and south pole. (This is, of course, a joke. You don't get two worms if you cut a worm in half. But you do get two magnets.)
- ❖ If you run a magnet a few times over an unmagnetized piece of a magnetic material (such as an iron nail), you can convert it into a magnet as well.

magnetic field. Then the flux is cut by the conductor and an EMF is induced.

The magnitude of EMF inducede is $e = BlV \sin\theta$

3.5.3 Statically Induced Emf

By keeping a conductor or coil in statically and varying the magnetic field will induce an EMF in the conductor or coil which is statically induced EMF.

Statically induced EMF can be classified as self inductance and mutual inductance.

(a) Self Induction

- This is the EMF induced in a coil due to the change of its own flux linked with it.
- If current through the coil is changed, then the flux linked with its own turns will also change, which will produce self induced EMF.
- The induced EMF is always opposite in direction to the applied EMF.

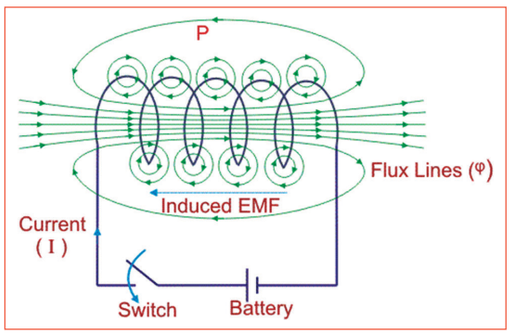


Fig 3.13 *Self Induction*

(b) Mutual Induction

It is the ability of one coil to produce an EMF by induction. When the current in the second coil changes, both coils are placed nearer.

- When two coils are placed nearer and current is passed through one of the coil, magnetic flux will be produced which is common to both coils.
- When current through first coil is varied, the magnetic flux will vary, which will induce an emf in second coil.

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